

The constant below the logarithm:

$$Z_2(n) = n^{-1/2} \left(\frac{A}{2} \log n + 2A \log b + C_0 \right) + O(n^{-1})$$

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Abstract

Unit 35 established the leading term $Z_2(n) \sim \frac{A}{2} n^{-1/2} \log n$ of the $d = 2$ chart integral with coinciding exponents. This unit identifies the next term exactly. The logarithmic primitive $H(L) = \int_0^L F(x)/x dx$ satisfies $H(L) = A \log L + C_0 + \theta(L)$ with $0 \leq \theta(L) \leq M/L$ for $L \geq 1$ and the explicit constant $C_0 = \int_0^1 F(x)/x dx - \int_1^\infty (A - F(x))/x dx$; consequently $Z_2(n) = n^{-1/2} (\frac{A}{2} \log n + 2A \log b + C_0) + O(n^{-1})$ with an explicit remainder, and $\sqrt{n} Z_2(n) - \frac{A}{2} \log n \rightarrow 2A \log b + C_0$. Zero sorry/axiom.

1 Setting

The squeeze $A \log L - M \leq H(L) \leq A \log L + E$ of unit 35 pins the $O(1)$ term only to a window. Since $A - F(x) = \int_x^\infty f \leq M/x$, the tail integrand $(A - F(x))/x$ is $O(x^{-2})$ and integrable on $[1, \infty)$, so the window collapses to a single constant with a rate: the difference $H(L) - A \log L - C_0$ is exactly the tail $\int_L^\infty (A - F)/x$, which is nonnegative and at most M/L .

2 The things formalised

```

noncomputable def logConst ( a : ) : :=
  ( x in Ioc 0 1, quadPrimitive a x / x)
  - x in Ioi 1, (quadMass a - quadPrimitive a x) / x
theorem logPrimitive_sub_log_sub_const ... (hL : 1 L) :
  logPrimitive a L - quadMass a * Real.log L - logConst a
  = x in Ioi L, (quadMass a - quadPrimitive a x) / x
theorem logPrimitive_sub_log_sub_const_bounds ... : 0 ... quadMoment a / L
theorem twoDIntegral_second_order_isBig0 ( a b : ) (h : 0 < ) (hb : 0 < b) :
  (fun n => twoDIntegral a b n - n ^ (-(1/2 : )) * twoDSecondOrderCoeff a b n)
  =0[atTop] fun n => n ^ 1
theorem twoDIntegral_second_order_tendsto ... :
  Tendsto (fun n => Real.sqrt n * twoDIntegral a b n - quadMass a / 2 * Real.log n)
  atTop ( (2 * quadMass a * Real.log b + logConst a))

```

3 Proof strategy

The existing `logPrimitive_split` already writes $H(L) - A \log L$ as $\int_0^1 F/x - \int_1^L (A - F)/x$; splitting $\int_1^\infty = \int_1^L + \int_L^\infty$ with `setIntegral_union` on $(1, L] \cup (L, \infty)$ gives the exact tail identity. The tail is compared to $M \int_L^\infty x^{-2} = M/L$ via `setIntegral_mono_on` and Mathlib's

`integral_Ioi_rpow_of_lt`. For Z_2 , the reduction $Z_2 = n^{-1/2}H(\sqrt{n}b^2)$ of unit 35 and $\log(\sqrt{n}b^2) = \frac{1}{2}\log n + 2\log b$ turn the bound into $0 \leq \sqrt{n}Z_2 - (\frac{A}{2}\log n + 2A\log b + C_0) \leq M/(b^2\sqrt{n})$ once $n \geq \max(1, b^{-4})$; the big- O and the limit are read off this two-sided estimate (the limit by `squeeze_zero`). The constant and the tail bound were checked numerically first (residual 10^{-16} at $L \geq 5$).

4 Roadblocks and resolutions

- `Continuous.const_sub` does not exist; write `continuous_const.sub h`.
- $(\sqrt{n})^{-1} \cdot (1/\sqrt{n}) = n^{-1}$: `field_simp` followed by rewriting $n = \sqrt{n}\sqrt{n}$ rewrites the n inside the radical too (a known trap). Isolate the step as `one_div, ← mul_inv, Real.mul_self_sqrt`.
- `squeeze_zero` cannot infer the dominating function from the later goals; pass `(g := ...)` explicitly.

5 What was Mathlib, what was new

Mathlib: `integrableOn_Ioi_rpow_of_lt`, `integral_Ioi_rpow_of_lt`, `setIntegral_union`, `Ioc_union_Ioi_eq_Ioi`, `setIntegral_mono_on`, `squeeze_zero`, `tendsto_inv_atTop_zero`.
 New: the constant C_0 of the logarithmic primitive with its sharp one-sided rate, and the two-term expansion of the $d = 2$ log integral.

6 Lessons

When an $O(1)$ window comes from two one-sided bounds, look for the exact tail identity behind it: here the “window” is literally a nonnegative tail integral, so the constant term and its rate cost one lemma each. The same mechanism will give the constant term below $(\log n)^{d-1}$ for the iterated primitives H_m .

7 Outcome

The $d = 2$ coinciding-exponent chart integral now has a two-term expansion with explicit constant and $O(n^{-1})$ remainder, complementing units 35–36 (leading term) and 40 (mixed multiplicity). Next candidates: the constant term for H_2 (the $\log n$ coefficient below $(\log n)^2$ at $d = 3$), or the weighted primitives F_γ for general (k, h) .